

Code No: 155AT**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, January/February - 2023****DATA ANALYTICS****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define Data Management. [2]
- b) Which source of Data is more reliable and Why? [3]
- c) What are the steps involved in Analysis of Data? [2]
- d) List out the features of Cloudera Impala. [3]
- e) Where can be Logistic Regression concepts applied? [2]
- f) Define Variable Rationalization. [3]
- g) Mention the applications of Time Series Analysis. [2]
- h) Name some Software to make Decision Tree. [3]
- i) What is Chernoff Faces technique? [2]
- j) What are the advantages of Data Visualization? [3]

PART – B**(50 Marks)**

- 2.a) What are the constraints and influences that will have an effect on Data Architecture Design? Explain.
- b) Discuss about the Data Reduction as a Data Preprocessing Step. [5+5]

OR

- 3.a) What is Secondary Data? Classify the Secondary Data Sources.
 - b) Discuss how data quality assessment can depend on the Intended Use of the data? [5+5]
- 4.a) List and explain the various Analytics techniques.
 - b) Explain the various methods to identify the Gaps in the Data and their handling mechanisms. [4+6]

OR

- 5.a) Discuss about the importance of Data Modeling in Business.
- b) What are the types of No SQL tools based on Data Models? Explain. [4+6]

- 6.a) What is meant by BLUE property? What are the blue properties of OLS method?
b) Discuss in detail about Multinomial Logistic Regression. [5+5]

OR

- 7.a) Explain the purpose of the Least Square Estimation in Regression with an example.
b) Discuss in short about the following Model Fit Statistics:
i) Hosmer Lemeshow Test ii) Error Matrix. [5+5]
- 8.a) What is Overfitting? How to Prevent Overfitting?
b) Compare and Contrast between ARMA and ARIMA. [5+5]

OR

- 9.a) Differentiate supervised and unsupervised learning.
b) Discuss the STL approach for Time Series Decomposition. [5+5]
- 10.a) Demonstrate Geometric projection visualization techniques.
b) List out various applications of data visualization. [5+5]

OR

- 11.a) Write down steps involved in Data Visualization In Tableau.
b) Explain about tag cloud visualization technique. [5+5]

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Code No: 155AT**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, July/August - 2023****DATA ANALYTICS****(Computer Science and Engineering - Networks)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define outliers. [2]
- b) Explain how data is extracted from signals? [3]
- c) What is the need for business modeling? [2]
- d) Discuss about missing imputations. [3]
- e) Define sigmoid function. [2]
- f) What is the meaning of epsilon (ϵ) in regression equation? [3]
- g) Define the concept of pruning. [2]
- h) Discuss briefly about Arima. [3]
- i) Define data visualization. [2]
- j) What is the role of Tableau in Data Visualization? [3]

PART – B**(50 Marks)**

2. Design Data Architecture and manage the Data for analysis. [10]
OR
3. Illustrate in detail about the concept of managing the data for analysis. [10]
4. Demonstrate the types of regression with the help of a neat diagram. [10]
OR
5. Give examples to explain the places where analytics is used. [10]
6. With the help of an example, explain the concept of linear regression. [10]
OR
7. Describe in detail about Logistic Regression and its components. [10]
8. Compare and contrast between regression and segmentation. [10]
OR
9. Give an example to show the process of feature extraction from generated model. [10]
10. Explain the geometric projection visualization technique with the help of an example. [10]
OR
11. Demonstrate the example to explain the concept of Hierarchical Visualization Techniques. [10]

Code No: 155AT**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech III Year I Semester Examinations, March - 2024****DATA ANALYTICS****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What do you mean by missing values? [2]
- b) Which source of data is more reliable? Why? [3]
- c) What is the objective of reporting? [2]
- d) What is the difference between nominal and ordinal? [3]
- e) What is residual? [2]
- f) How to compute cost function in linear regression? [3]
- g) What is supervised learning? [2]
- h) What is the use of gini index? [3]
- i) What is data visualization? [2]
- j) What are the benefits of using stick figures? [3]

PART – B**(50 Marks)**

2. Explain the different steps in data processing. [10]
- OR**
3. How to handle noise and outliers in data management? Explain with examples. [10]
4. Provide an overview of data modeling techniques and their application in business. [10]
- OR**
5. Describe different tools and environments for performing data analytics. [10]
6. Demonstrate ordinary least squares method for linear regression. [10]
- OR**
7. Make a comparison of linear regression with logistic regression. [10]
8. What is a decision tree? Explain CHAID algorithm for building decision trees. [10]
- OR**
- 9.a) Cleaning in ETL performs basic data unification. List the rules associated with this process.
- b) Discuss the measures for time series data forecast accuracy. [5+5]
10. Explain geometric projection visualization techniques. [10]
- OR**
11. Discuss the challenges involved in visualizing complex data and suggest suitable techniques. [10]

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech III Year I Semester Examinations, August/September - 2024

DATA ANALYTICS

(Common to CSE, IT)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

- 1.a) What is an outlier? [2]
- b) What is null hypothesis? [3]
- c) What are NoSQL databases? [2]
- d) What do you mean by “feature similarity”? [3]
- e) Define the term “regression”. [2]
- f) What is root mean square error? [3]
- g) What is the difference between data warehousing and data mining? [2]
- h) Define the core idea of principle component analysis. [3]
- i) List any two limitations of data visualization techniques. [2]
- j) Mention any three alternatives for word clouds to visualize the text data. [3]

PART – B

(50 Marks)

2. Outline several sources of data for data collection and compare those sources of data with advantages and limitations. [10]

OR

3. What is the importance of data processing? Describe the steps involved in data processing. [10]

- 4.a) Discuss about the types of data variables.
- b) Write about the application of modeling a business and need for business modeling. [5+5]

OR

5. Summarize different types of data models with suitable examples. [10]

6. Outline neatly the purpose of ‘least square estimation in regression’ with example. [10]

OR

7. “Logistic regression is an example of non-linear regression” prove the statement with appropriate case study and experiment. [10]

8. Explain the following time series models with examples.

a) Auto Regression (AR)

b) Auto Regressive Moving Average (ARMA).

[5+5]

OR

- 9.a) What is STL approach? Explain in detail.
b) Discuss about supervised and unsupervised learning basics. [5+5]

10. Infer about “Geometric projection visualization techniques” in detail with case study. [10]

OR

11. Outline the concept of “Icon-Based Visualization Techniques” with case study. [10]

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